Design by Falsification:
an Evidence-Bound Multi-Agent Harness Distilled from a Council of Experts That Mostly Did Not Work

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Abstract

We set out to build a council of experts: domain-specialist language models feeding a lead writer, on the premise that specialists know things generalists do not, that instructions can make the lead careful, and that models can tell you when they are out of their depth. Across more than fifty pre-registered experiment cells on auditable local hardware (7–20B models), nearly every premise was falsified — specialist fine-tunes carry per-model, not per-domain, signatures; an epistemic instruction installs only its named phrase; preference judges then penalize that phrase; models' stable self-reports (confidence, priors, ownership) do not predict their behavior; and prose transports only 0.11–0.33 of the epistemic qualification it is given. This paper reports what happened next. The falsifications were themselves findings, and they compose into an architecture: an orchestration harness in which redundancy is engineered rather than hoped for, epistemic freight travels outside the writer's prose, re-consultation is triggered by mandated artifacts rather than self-assessment, and no component survives on rationale alone. Every load-bearing design rule was then re-tested prospectively as its own registered cell. Assigning a contested quantity to a second specialist halves corrupted-value adoption (0.900 → 0.450); an appendix assembled outside the writer carries 100% of specialist caveats against the 17% that survives prose, flips downstream decisions (+0.691 over a matched irrelevant control) at parity with in-prose placement, and shows no detected preference cost; and the full re-consultation loop closes with a live specialist (0.905 adoption vs. 0.333 control, within noise of a scripted ceiling). Two registered tests went *against* our design rationales — sibling isolation lost its contamination warrant to an informative null, and an elicited ownership law died in its confirmation cell at exactly chance — and the harness records both demotions in place. We argue this is the correct acquisition path for multi-agent architecture: not accumulating plausible components, but keeping only what survives registered attempts to kill it.

1 Introduction: the council that was supposed to work

The plan was conventional wisdom circa 2024–2025. Take three domain-specialist models — healthcare, legal, finance — have each analyze a decision problem from its own expertise, and let a lead writer synthesize their contributions. Aggregation pipelines of this shape report state-of-the-art preference results [1]; specialist fine-tunes are abundant; and the architecture's premises feel almost too obvious to test: specialists contribute what generalists cannot; instructing the lead to preserve uncertainty makes it careful; a model that is out of its depth can say so; corroboration among specialists strengthens claims.

We tested them. The program ran as a falsification-first ledger: every cell pre-registered with predictions, falsification conditions, and attainability computed from the archive before any run; every deviation and verdict recorded; instruments themselves audited (fourteen recorded instrument-validity findings). A companion paper [2] reports the audit findings in full. This paper tells the systems story: **what it looks like to lose most of your architecture's premises in public, and what you can build from the parts that survive**.

The arc, compressed:

1. **The premises died** (§2). Not to opinion — to registered cells with pre-stated kill conditions, most of which fired.

2. **The corpses were load-bearing** (§3). Each falsification localized a real mechanism: the lead is a source selector, not a verifier; compliance is not behavior; the scarce resource in consultation is the trigger, never the capability.

3. **The mechanisms compose into a harness** (§4). If protection is coverage, engineer coverage. If prose loses freight, do not ship freight in prose. If self-assessment is empty, trigger from mandated artifacts instead.

4. **The harness was then attacked, component by component** (§5). Six registered verdicts: four components survived with causal or predictive effect sizes, one design rationale was demoted by an informative null, and one candidate law was killed by its own confirmation cell. The architecture that remains is smaller than what we designed and better than what we assumed.

Why this is a systems contribution. Multi-agent frameworks ship components on rationale: a verification stage because verification sounds prudent, a confidence threshold because calibration sounds measurable, a debate round because deliberation sounds epistemic. Our central claim is methodological: **every component of a multi-agent system is an empirical claim about model behavior, and most such claims are false at the scale where practitioners actually deploy them.** The harness presented here is not offered as the best possible architecture. It is offered as the subset of one architecture that survived fifty-plus registered attempts at its premises, with the verdicts — including the demotions — attached to the components they license.

2 How the premises died

Table 1 is the program's obituary column. Each row is a premise the council needed, the registered cell that tested it, and what happened. Effect sizes and protocols are in the companion paper; here we give each one sentence of mechanism.

Three of these deserve a sentence more, because the harness is built directly on their corpses.

Compliance is not behavior. The phrase-swap design put two synonymous instruction forms against a control whose bytes differ only in the instructed clause. The instruction reliably bought its own phrase and nothing measurable beyond it — and a dictated-phrase registry with a measured zero over-attribution rate (0/2,907 spans) guarantees the accounting. Every "be careful"-style component a framework ships is, on this evidence, buying surface forms.

The lead is a source selector. With a clean co-source covering a contested quantity, corruption never propagated (0/27); with no clean source, it propagated or the quantity was

premise the council needed	test	verdict
Specialist fine-tunes share a domain signature	two medical tunes of one base vs. their base	**falsified** — signatures are per-model; the tunes disagree with each other
An epistemic instruction changes the lead's behavior	counterbalanced phrase-swap; synonym forms	**falsified** — the named phrase appears (0.67–0.70); the construct beyond it is unchanged (CIs $[-0.079, +0.111]$, $[-0.103, +0.087]$)
Careful marking is at worst neutral under evaluation	order-debiased preference, phrase arms vs. own control	**reversed** — the phrase *loses* preference (0.261, 0.317; replicated cross-family)
A specialist can flag its own limits	roster + deferral token; in- vs. out-of-domain	**falsified** — routing is accurate when fired (0.909) but discrimination spans zero (0.306 vs. 0.083)
Corroboration among specialists strengthens transport	planted agreement across contributions	**null** — 0.487 vs. 0.524, CI spans zero
The lead verifies or recomputes	corrupted-source injection	**falsified** — it *selects* sources: 0/27 propagation with a clean co-source, 5/27 when all sources corrupt
The council improves accuracy	verifiable-outcome battery	**no evidence** — diff CI $[-0.085, +0.034]$; held as absence of evidence, with the harder battery still owed
Prose carries the specialists' epistemic content	sentence-level transport, two writers, two corpora	**bounded** — slopes $w = 0.11$–0.33; two thirds to nine tenths of qualification dies in synthesis
Models' elicited self-state predicts behavior	priors (0.544); elicited ownership map	**falsified twice** — the ownership map was stable and near-unanimous, and predicted arbitration at exactly 0.500 against a descriptive 0.895

Table 1: The premises, their registered tests, and their verdicts. Full protocols, CIs, and preregistrations in the companion paper and the public ledger.

silently dropped. The lead does not check; it *chooses*. This single mechanism sets both the ceiling (no verification will emerge from prompting) and the opportunity (choice can be engineered by controlling what sources exist).

The trigger is the scarce resource. In every consultation design we measured, the mechanism worked when fired and the firing was rare: the lead uses a routed clarification 7/7 but names the tension that warrants it at 0.34; specialists route correctly at 0.909 but recognize the need at 0.306; across three cells the trigger rate ranged 0.208–0.583 by batch. Self-assessment, the faculty frameworks lean on for escalation, was the one faculty we never found.

3 What survived

Three measured gains outlived the audit, and they are the only three the harness claims.

Aggregation wins preference. The pipeline beats the same writer answering directly in 0.818 [0.718, 0.919] of decisive order-debiased pairs, surviving length matching and replicating under a disjoint judge family (0.788). The composition of this lift resists our full surface-feature set and is reported as unidentified — but it is real, and it is not the epistemic behaviors the prompts request.

Error immunity where coverage overlaps. The source-selector mechanism, read constructively: 0/27 propagation when an uncorrupted co-source exists.

Routed re-consultation resolves contested quantities. When an orchestrator dispatches a follow-up and the reply is informative, the lead adopts the implied resolution at 1.000 (scripted) and 0.905 (live specialist) against 0.333 without the loop — with a length-matched contentless reply landing at 0.200, so the information does the work, not the ritual.

4 The harness: falsifications, composed

Each component below is derived from a specific verdict in §2, cites it, and does only what that verdict licenses. Figure 1 shows the assembled pipeline with every edge labeled by its measured number. The full architecture document ships with the ledger.

Seating is empirical, never categorical. Because signatures are per-model, no seat is chaired for its category label. A candidate passes a measured gate: a degeneracy screen (≥ 800 visible characters on screen items — one archive fine-tune produced 12/12 degenerate probe runs), a tuned-minus-own-base signature probe, and a format smoke test at production token budgets. A generalist under a role prompt is a legitimate seat if it passes; most of our cells ran on exactly that.

Isolation, on cost grounds only. Seats receive their task and a one-line roster. Sibling outputs are not forwarded — but the original contamination rationale is dead (§5); the rule survives because sibling context costs $\sim$10k input characters and 14% longer outputs per seat while buying no measured change in lane discipline.

Redundancy is engineered. The orchestrator identifies load-bearing quantities in the decomposition and assigns each to at least two seats. This is where the council's error immunity lives; it does not exist by default.

Triggers come from mandated artifacts. The lead runs two-stage: a mandated tension list first (391/396 archived runs produce one when required), synthesis second. The orchestrator — never any model's disposition or self-assessment — parses the list and dispatches follow-ups. A reply that adds nothing beyond the seat's first contribution is dropped, not delivered (a ritual-effect risk observed at $d \approx 0.22$, below our power to confirm, is treated as live).

Epistemic freight travels out-of-band. Caveats, assumptions, and confidence never route through the writer: seats emit them as structured fields and the harness carries them mechanically to an appendix of the final artifact. The writer receives *no* disposition instructions at all — the evidence says they buy nothing and their surface costs preference.

A Goodhart charter. Trigger rates, hedge counts, compliance shares, transport slopes, and preference scores on epistemic content are diagnostics and may never be optimization targets; the optimizable signal must live outside the pipeline's own text. Each entry on the forbidden list cites the cell that showed what optimizing it would manufacture.

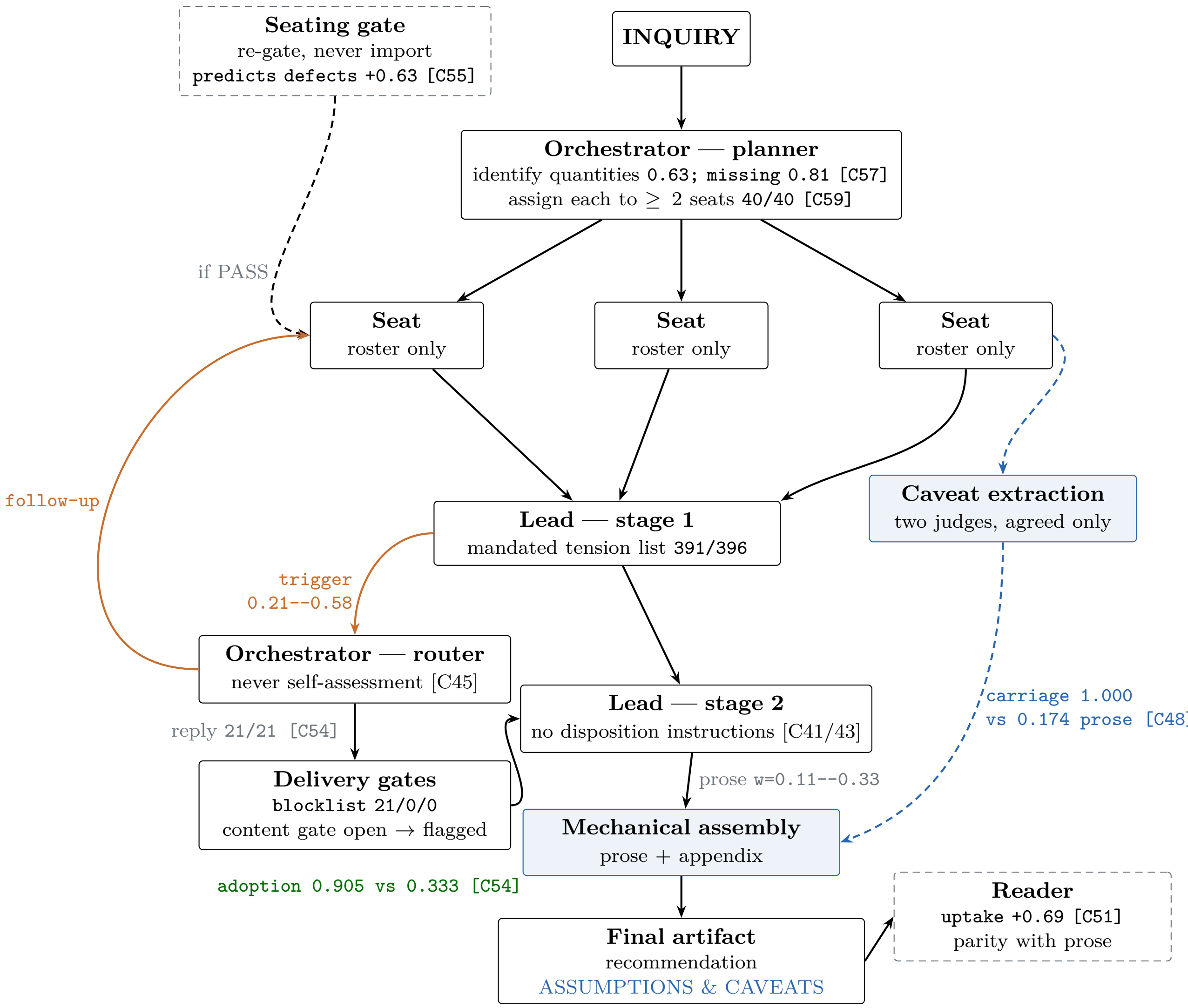

Figure 1: The assembled harness on one inquiry, every edge labeled by its registered-cell measurement. Dashed blue: the out-of-band freight lane the writer never sees. Orange: the orchestrator-routed re-consultation loop, bounded by its measured trigger rate. Seats are interchangeable by industry — chairs are earned by the gate, never by category label — and engineered overlap between them is where the error immunity lives: corrupted-value adoption 0.90 → 0.45 [C47], the clean co-source used end-to-end at +0.43 [C59].

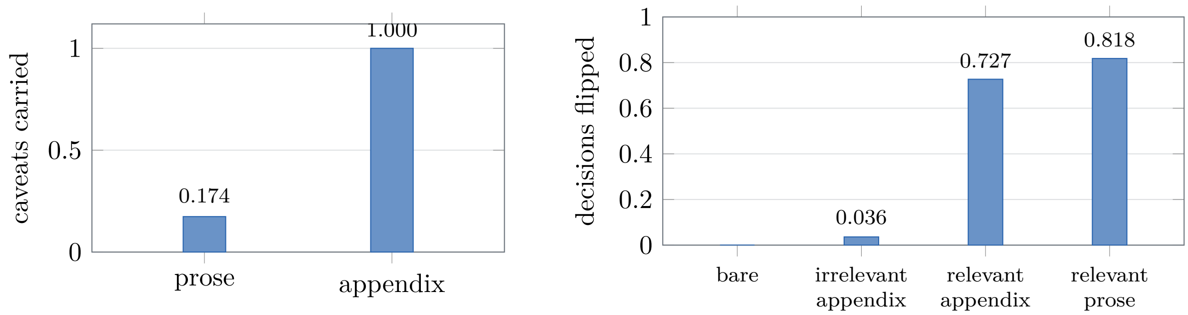

Figure 2: The out-of-band channel, attacked on both halves and surviving. *Left*: caveat carriage through writer prose vs. the mechanically assembled appendix (zero-generation design). *Right*: registered two-token reader decisions flipped by a planted decision-relevant caveat, by delivery arm; the irrelevant-appendix control shows the effect is content, not appendix-presence. Uptake +0.691 [+0.455, +0.891]; parity with prose +0.091 [−0.036, +0.218].

5 The harness attacked: prospective re-tests

Design rules distilled from findings can still be wrong — post-hoc patterns calcify. So each load-bearing rule got its own registered cell, with kill conditions stated first. Six verdicts follow.

The seating gate predicts (survived). Ten local candidates were gated on the frozen degeneracy screen — two screen items disjoint from the pipeline case set — with verdicts committed to the repository *before* any pipeline run, then every candidate was seated on the pipeline cases. Strict rank separation held: both gate-fail candidates defected in the pipeline (0.333 and 0.944 of contributions) while all eight gate-pass candidates ran at 0.000–0.056; the pooled contrast is +0.632 [+0.549, +0.722]. Two refinements rode along: gate verdicts *age* (one model that passed the archive-era screen failed the fresh re-gate, then defected in-pipeline at 0.333 — seating must re-gate, never import a stored verdict), and format compliance is a separate axis that does not predict seat defects (the legal fine-tune failed the format smoke 0/4 yet seated defect-free; that component's proper target is verdict-emitting roles).

Redundancy is causal (survived). One factor — assign the contested quantity to a second seat — halved corrupted-value adoption, 0.900 → 0.450 ([+0.200, +0.725]), with the clean value actually adopted (0.000 → 0.500): selection, not dilution. The bare arm's 0.900 doubles as the sharpest no-verification number on record. Protection is bimodal across items; two candidate boundary mechanisms were then registered and killed (prior-plausibility at 0.544; the elicited ownership map at exactly 0.500 prospective against 0.895 descriptive, with an MDD of 0.68 — the confirmation cell doing its anti-calcification job). The boundary is OPEN and stated as such.

Out-of-band freight carries and is used (survived). Carriage: the appendix delivers 1.000 of both-judge caveat sentences against 0.174 surviving prose, with no detected preference penalty (0.510 [0.267, 0.754]; not evaluable at power, reported as bounded). Uptake: appendix-delivered caveats flip a reader's registered decision at 0.727 against 0.036 for the identical block carrying an irrelevant caveat, with direction-balanced items so caution cannot masquerade as uptake, at parity with in-prose placement (Figure 2). The channel preference cannot see is not an exhibit; it changes decisions.

The live loop closes (survived, with one halt en route). A first live-specialist cell buried the deciding fact in the specialist's own round-one text and *halted at its registered pilot* *gate*: the trigger named the tension at only 0.208 with the fact visible in the pile. The redesign supplied the fact as the specialist's private working notes — knowledge held but never written — and restored the original round-one bytes. The live specialist conveyed the fact 21/21 when dispatched; the lead adopted at 0.905 vs. 0.333 archived control (+0.571 [+0.139, +0.912]), within −0.095 [−0.227, 0.000] of the scripted ceiling. One new risk is on record: 4/21 live replies embellished the correctly conveyed fact with *fabricated* authority — invented case law, invented regulatory rulings — and were adopted anyway. Content gates screen absence, not invention; the fabrication rate is observed, not bounded.

Isolation lost its warrant (demoted). The registered one-factor test made byte-fixed sibling contributions visible under production prompts: off-domain framework density moved −0.009 [−0.048, +0.028] against a pre-registered MDD of +0.05–0.08 computed from the archive — an informative null, with in-domain density unchanged. The contamination rationale is withdrawn; the rule stands on cost. (A registered secondary even pointed the other way: seats shown their siblings' coverage used slightly *fewer* off-domain terms of their own, −0.033 [−0.053, −0.013] — recorded as a candidate mechanism, not claimed.)

The ownership law died in confirmation (killed). A stable, near-unanimous elicited map of which seat "owns" each quantity — exactly the kind of pattern a framework would ship as a routing table — predicted arbitration at chance on fresh items. We report this kill with the survivals because it is the argument: the difference between an evidence-bound harness and a prompt stack is that the harness can lose these arguments in public, on the record, component by component.

6 Operating economics and open risks

The trigger bounds coverage. Everything downstream of the trigger is now measured and works; the trigger itself fires at 0.208–0.583 by batch and is deliberately not an optimization target (raising it blind manufactures the ritual condition the loop's own control arm exposed). Deployments should budget coverage accordingly: the loop resolves the contested quantities it notices, and it does not notice most of them.

Open risks, stated. Per-item protection is bimodal with the boundary mechanism unknown (two candidates killed). Live-reply fabrication is now *bounded* by a registered follow-up cell: reply-level 0.100 [0.047, 0.201] with the deciding fact in hand, 0.150 [0.081, 0.261] without it — concentrated entirely on one trigger topic (15/20 there, 0/100 elsewhere) and name-stable, the same two invented decisions recurring across independent samples, which licenses a block-list delivery filter; the seat never admits the gap (0/20 sampled), the backfill mechanism is directionally positive but not evaluable at six clusters, and the bound covers adversarial case citations only — a floor, not a ceiling, on confabulation. The preference cost of the appendix is bounded-not-evaluable at current power. The council's accuracy question is now RESOLVED as untestable for lack of headroom: two registered batteries (multi-step computation; rule-path selection) put the direct writer at ceiling (0.977, 0.983) under baseline-only difficulty screens, so no well-specified known-answer battery can measure a council difference at this scale — the strongest available statement of the design rule that a council is not an accuracy device. None of these is hidden in the architecture document; each is attached to the component it concerns.

Running the harness replicates the science. The harness's telemetry doubles as a replication battery: every deployment yields transport estimates, trigger telemetry, phrase-swap slots for proposed instructions, and preference-vs-verified divergence wherever ground truth exists — each with its attainability gate. An honest deployment is also, at near-zero marginal cost, the external validation the program's scope caveats call for.

7 Related work

MoA [1] and successors report preference gains from aggregation; Self-MoA [3] shows mixing weaker models often hurts, consistent with our per-model signature result. Debate and multi-agent-discussion results [7] find gains concentrated where prompts were weak, consistent with our finding that the one working intervention routes *information*, not deliberation. LLM-judge critiques [4] motivate our order-debiased two-family protocol; length and sycophancy pressure [5, 6] sharpen to our measured phrase penalty. We are not aware of prior work that pre-registers each component of a multi-agent architecture and reports the failures with the survivals; that practice, more than any single component, is what we propose for adoption.

8 Limitations

Scale and scope: 7–20B local models, advisory-domain English, one primary writer family (cross-family replications where stated). Cluster ceilings of 6–18 items bind every harness cell; the live-loop comparison uses archived arms under a registered fresh-batch check; the uptake verdict covers 11 of 18 authored items after a registered floor gate; trigger rates are batch-variable and quoted only as a range. The harness's gains are preference, immunity, and resolution — *not* accuracy, which remains untested at the standard the program set for itself. Verdicts inherit their cells' scope and should not be generalized past it; the architecture document states, for every component, exactly which cell licenses it.

9 Reproducibility

The complete pre-registration ledger (52 registration blocks at this writing), every deviation and verdict including the two halts, the architecture document, the measurement kit (`gst`), the frozen dictation registry, and all item files are in the project repository; registrations are committed before first runs, so temporal ordering is verifiable from history.

References

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